

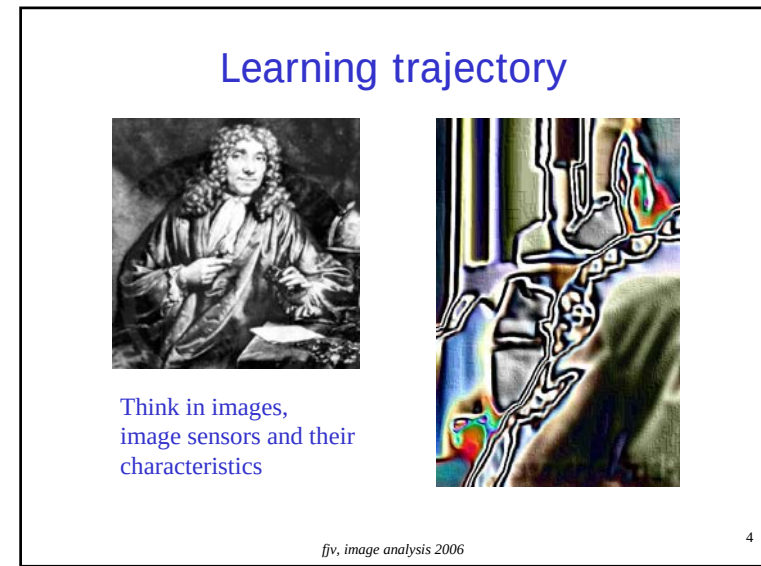
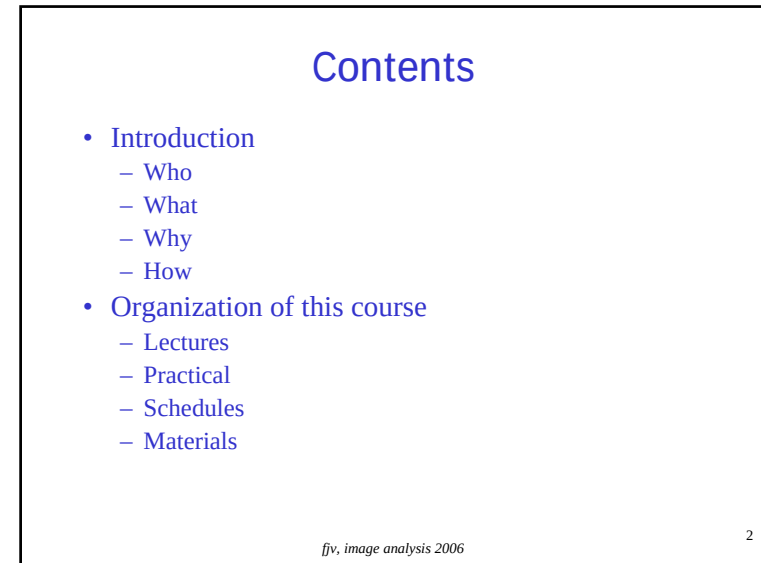
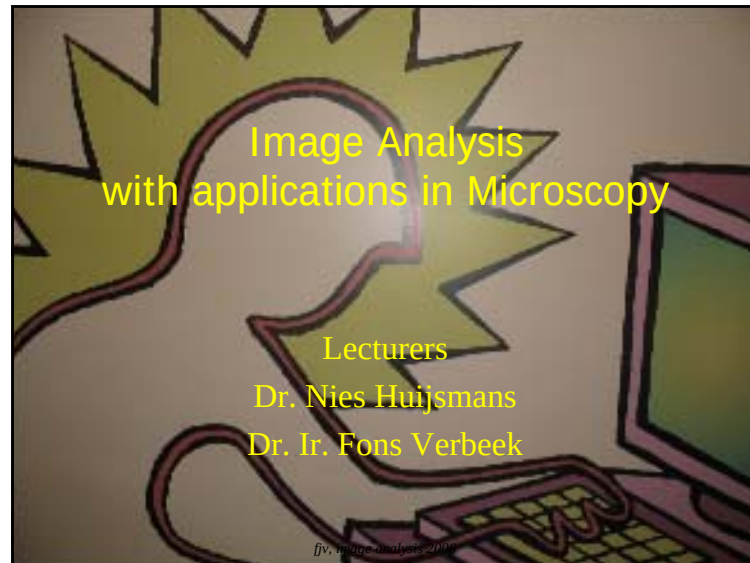


Image Analysis

- Image analysis is probably what we are doing for the larger part of the day.
- Capturing images:
 - 1839 Daguerre (positive on silvered plate).
 - Colour: Lumiere ~ 1900 (Autochrome)
- Vision, Computer Vision
 - Computer graphics, digital images
 - ~1980 specialized hardware for Video
 - ~1990 improvement CCD sensors, Expensive Hardware
 - ~2000 CCD public market, Multimedia PC's

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Microscopy & Analysis



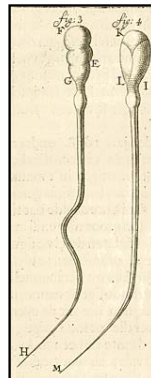
Antoni van Leeuwenhoek (1623 – 1723)



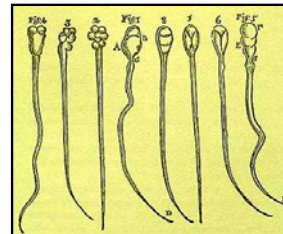
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Imagine - Imaging



- Spermatozoa
- Little animals?, Cell
- Reproduction

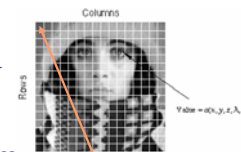


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Digital Image

- Computer: how to represent an analogue image in memory
- Digital Image
 - Characteristic: it is **sampled** from a continuous signal to a discrete signal
 - Discrete units can be represented
 - Every discrete unit is a sample point
 - Sample point in a picture = **picture element**
 - Picture element = PIXEL
 - Image element = IMEL



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Issues of Digital Image

- What is the size of a PIXEL, IMEL
 - The size is determined by lens and sensor
 - Resolution, what is the smallest we can see
 - Measurement & Analysis
- What is the dynamic range of a PIXEL
 - Density
 - How many values to represent density
 - 8-bit (byte) Grey Value, 24-bit Color
- What does the dynamic range mean
 - Reflection
 - Transmission
 - Fluorescence

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Resolution

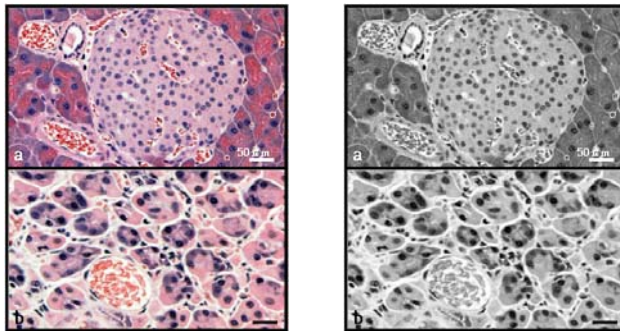


Gala Contemplating the Mediterranean Sea,
(Homage to Rothko), 1976: Figueras

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Transmission



24 bit color, transmission

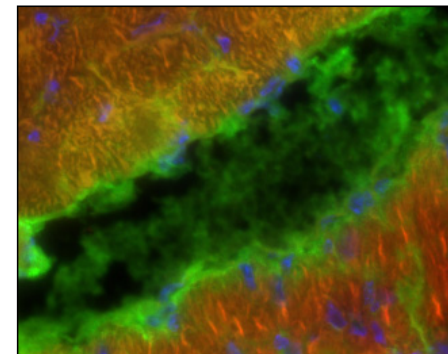
8 bit b-w, transmission

Black = no transmission

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Emission: Fluorescence

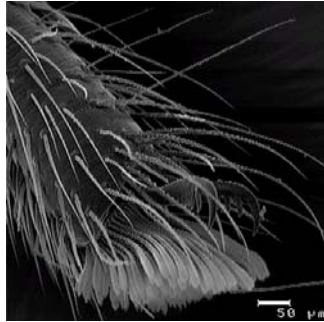


Emission of light. Black = no emission

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Reflection +/- light



SEM, jump spider foot, micro-van der Waals forces.

(Kessel et al., 2004)



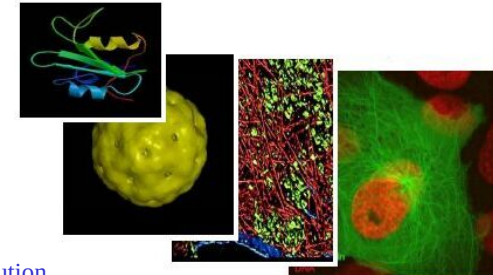
Stereo, zebrafish 24 hrs. gene expression

(Welten et al., 2006)

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Cyttron, a virtual microscope



- atomic resolution
- molecular resolution
- nanoscopic resolution
- microscopic resolution.

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A view on Images

- Image Processing:
 - Image in → Image out
- Picture Processing:
 - Explicitly to enhance the photographic output
- Image Analysis:
 - Image in → Numbers out
- Visualization, Computer Graphics
 - Numbers in → Image out

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Modern Approach Digital Imaging

- Consider computerized processes
 - Low-, Mid-, & High-level processes.
 - Low: image enhancement (image in, image out)
 - Mid: reduction (image in, attributes out)
 - High: Meaning (image analysis – computer vision)
- Computer Vision: Human Vision Emulation
 - Also including cognitive processes
- Image Analysis: Understanding image content by measures and numbers.

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Images are Signals

- Electronic Imaging
 - Image is a signal produced by a sensor
 - Readout of the sensor
 - Discretization of signal
- Signal processing
 - Paradigms
 - Sampling, Reconstruction
- Measurement
 - Features and Attributes in machine learning
 - First extract features (data reduction)

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Course: Acquisition

- Some historical background
- Physics of Light & Color Spaces
- Human Visual System & Computers
- Imaging and sampling
- Image acquisition
- Microscopy
- Microscopic images in life sciences
- Staining methods, contrast

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Course: Image Processing

- Histogram manipulation
- Filters, Neighborhood filters
- Fourier Transforms
- Non linear filters, mathematical morphology
- Connectivity, grids, metrics
- Coordinate transformations
- Image transforms and coding

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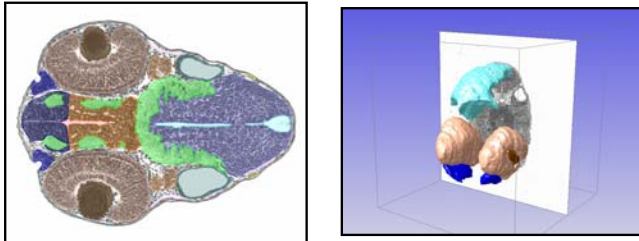
Course: Image Analysis

- Segmentation, Model based segmentation
- Edge detection
- Labeling
- Image features, measurement
- Pattern recognition, Clustering
- 3D image processing
- 3D reconstruction, modeling
- 2D, t analysis and/or 3D, t analysis

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From Image to Model



- Labeling and measuring biomedical images
 - Calibration
 - Registration

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Course Organization

- Theoretical part
 - Lectures, 13 x 2 hrs.
 - Lectures are on Thursday, 9.00 – 11.00, 312
- Practical part
 - Hands on Microscope, Image Acquisition
 - Assignments Image Processing/Analysis
 - Learning to work with Image Processing
 - Report

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Persons Involved

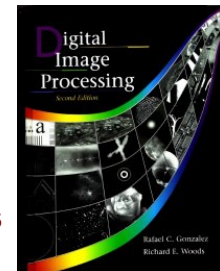
- Theoretical part
 - Nies Huijsmans
 - Fons Verbeek (FJV)
- Practical part, Laboratory exercises
 - Image acquisition & microscopy
 - Gerda Lamers & Fons Verbeek
 - Image processing/analysis assignments
 - Mounia Belmamoune, Yun Bei, FJV

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Materials Involved

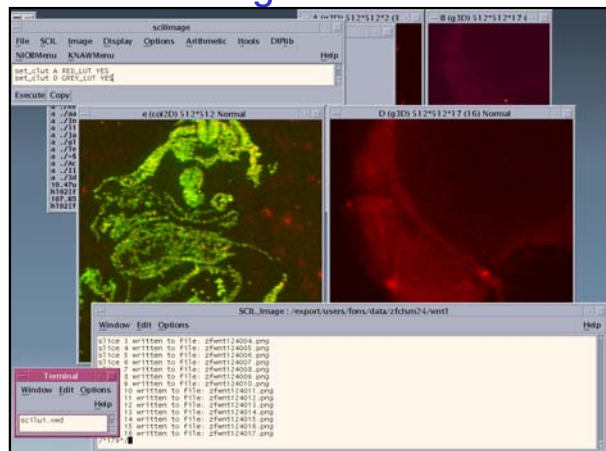
- Book (recommended)
 - Digital Image Processing, Rafael C. Gonzales, Richard E. Woods. Addison Wesley, 2000
 - Digital Image Processing using Matlab Rafael C. Gonzales, Richard E. Woods & Steven L. Eddins Pearson Prentice Hall, 2004
- Lecture notes
 - 13 lectures-notes available as pdf
 - Additional papers
- Software
 - Installed on LIACS systems
 - manuals
- Website:
 - <http://www.liacs.nl/~fverbeek/courses/ia2006>
 - News-section, Documents section



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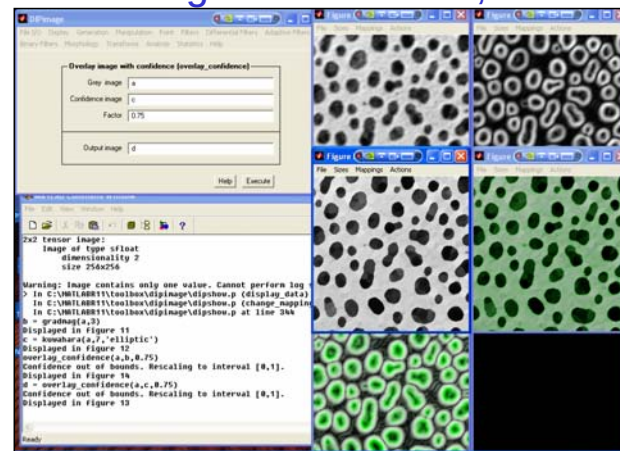
SCIL-Image Environment



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DIP-Image Environment, Matlab



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